

SMART ZEBRA CROSSING SYSTEM WITH EMERGENCY TRAFFIC CONTROL

RISHIKA R 23BMH1050

CHRISTA DARYN P 23BMH1126

1. Introduction

Traffic management and pedestrian safety are two major challenges in modern urban environments. With the rapid increase in vehicles on roads, conventional traffic signals that operate on fixed timing often fail to adapt to real-time traffic conditions. This leads to unnecessary waiting time for vehicles and unsafe conditions for pedestrians trying to cross roads. Zebra crossings are designed to allow pedestrians to cross safely, but in many cases vehicles do not stop appropriately because traditional systems lack intelligent monitoring and control.

Recent advancements in artificial intelligence and computer vision have made it possible to develop smart transportation systems capable of analyzing road conditions in real time. Object detection algorithms can identify pedestrians, vehicles, and other objects using camera inputs and make decisions accordingly. Integrating such intelligent systems with embedded hardware platforms enables automated traffic control mechanisms that respond dynamically to changing conditions.

The Smart Zebra Crossing System with Emergency Traffic Control is designed to address these issues by integrating machine vision, decision-making algorithms, and embedded hardware control. The system uses a camera to capture live video from the crossing area and processes the video using the YOLO (You Only Look Once) object detection model. Based on the detected objects, the system determines whether pedestrians or vehicles should be given priority. Additionally, the system can detect potential emergency vehicles and immediately grant them traffic priority.

The decisions generated by the computer vision system are transmitted to a Raspberry Pi using socket communication. The Raspberry Pi then controls traffic signals represented by LEDs through its GPIO pins. This creates a dynamic traffic control system capable of adjusting signal states in real time. The project demonstrates how artificial intelligence

and embedded systems can be combined to create efficient and intelligent transportation solutions suitable for smart city environments.

2. Aim of the Project

The aim of this project is to design and implement an intelligent zebra crossing system that improves pedestrian safety and traffic efficiency by using real-time object detection and automated traffic signal control.

3. Objectives

The main objectives of the project include:

1. To implement a real-time pedestrian and vehicle detection system using a deep learning object detection model.
2. To identify pedestrians present within the zebra crossing region using region-based analysis.
3. To count vehicles approaching the crossing area and analyze traffic density.
4. To develop decision-making logic that determines whether pedestrian or vehicle traffic should be prioritized.
5. To detect potential emergency vehicles and provide them immediate traffic priority.
6. To establish communication between the detection system and the hardware controller using socket programming.
7. To interface Raspberry Pi with traffic signal indicators using GPIO pins.
8. To simulate real-world traffic signal operations using LEDs representing vehicle and pedestrian signals.

4. System Architecture

The proposed system integrates computer vision, decision algorithms, and embedded control. The architecture consists of several components working together to perform detection, communication, and signal control.

The system begins with a camera that continuously captures video frames of the zebra crossing area. These frames are processed by a computer running the YOLO object detection model. YOLO analyzes the image and identifies objects such as pedestrians,

cars, buses, trucks, and motorbikes. The system then applies decision logic to determine whether pedestrian traffic or vehicle traffic should be given priority.

Once a decision is made, the system sends a command message through a network socket to a Raspberry Pi device. The Raspberry Pi acts as a hardware controller and receives these commands. Based on the received message, the Raspberry Pi activates specific GPIO pins connected to LEDs that simulate traffic signals.

Thus, the entire system can be represented as a pipeline consisting of image acquisition, object detection, decision making, communication, and hardware control.

5. Components Used

Hardware Components

The hardware components used in the project include a Raspberry Pi microcomputer, which serves as the main controller for the traffic signals. The Raspberry Pi provides General Purpose Input Output (GPIO) pins that allow direct control of electronic components.

LED lights are used to simulate traffic signals. Separate LEDs represent vehicle green, vehicle red, pedestrian green, and pedestrian red signals. Each LED is connected through a current-limiting resistor to prevent damage due to excessive current.

A breadboard is used to assemble the circuit connections without soldering. Jumper wires are used to connect the LEDs and resistors to the Raspberry Pi GPIO pins. A USB webcam is used to capture video input for the object detection system.

Software Components

The software implementation is developed primarily using the Python programming language. The system utilizes the OpenCV library for video capture and frame processing. The YOLOv8 object detection model from the Ultralytics library is used for real-time object recognition.

Socket programming in Python is used to establish communication between the laptop running the detection system and the Raspberry Pi controlling the hardware. The RPi.GPIO library is used to control the Raspberry Pi GPIO pins and manage the LED signals.

6. Working Principle

The working of the system can be divided into several stages.

Video Capture

The system begins by capturing video from a webcam placed near the zebra crossing area. The camera continuously records frames of the road environment where pedestrians and vehicles are present.

Object Detection

Each video frame is processed using the YOLOv8 object detection model. YOLO analyzes the image and detects various objects based on a pretrained dataset. The system specifically monitors objects classified as "person," "car," "bus," "truck," and "motorbike."

The detection model provides bounding boxes around detected objects along with confidence scores indicating the reliability of the detection.

Region of Interest Analysis

To ensure that only relevant pedestrians are considered, a Region of Interest (ROI) is defined within the frame. This region corresponds to the zebra crossing area. Only pedestrians detected within this region are counted for crossing decisions.

This approach prevents pedestrians who are far from the crossing area from affecting the signal control logic.

Decision Logic

The system applies a decision-making algorithm based on the detected objects. The algorithm counts the number of pedestrians in the crossing area and the number of vehicles approaching the road.

If an emergency vehicle is detected, the system immediately assigns priority to vehicles by activating the vehicle green signal.

If the number of pedestrians is greater than the number of vehicles, the pedestrian signal is activated to allow safe crossing.

Otherwise, vehicle traffic is allowed to continue by activating the vehicle green signal.

Communication

Once a decision is made, the system sends a text message representing the signal state to the Raspberry Pi using socket communication. The messages include "PERSON," "VEHICLE," or "EMERGENCY."

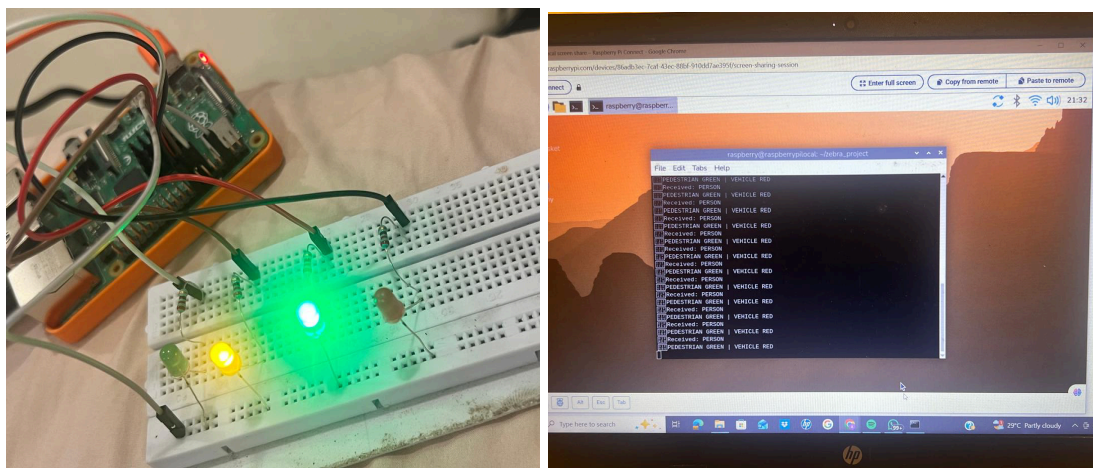
This communication allows the computer vision system and the hardware controller to operate independently while remaining synchronized.

Hardware Control

The Raspberry Pi continuously listens for incoming messages from the detection system. When a message is received, the Raspberry Pi activates the corresponding GPIO pins connected to the LEDs.

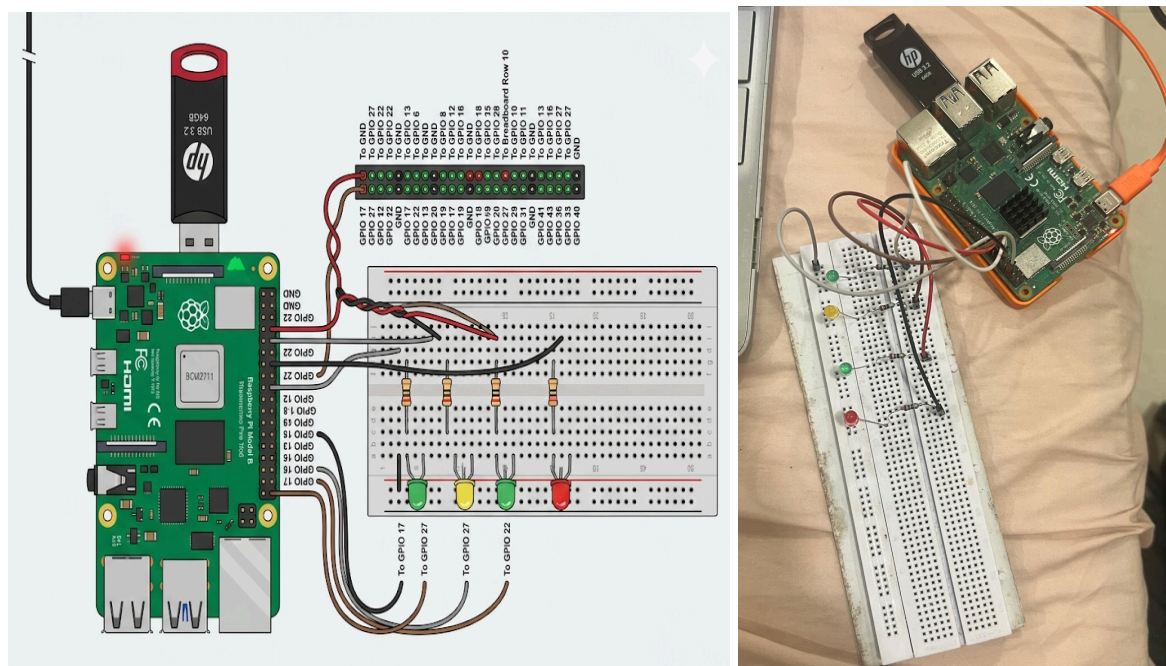
If a pedestrian crossing is required, the pedestrian green LED is turned on and the vehicle red LED is activated. If vehicle traffic is prioritized, the vehicle green LED is turned on and the pedestrian red LED is activated. In emergency mode, the vehicle green LED remains active to allow emergency vehicles to pass without delay.

Here vehicle red is yellow LED for differentiation purpose



The GPIO pins used in the project include GPIO17, GPIO27, GPIO22, and GPIO23.

In this prototype, a simplified traffic signal representation is used for demonstration purposes. The vehicle side utilizes **green and yellow LEDs** to indicate movement and caution states, while the pedestrian side uses **red and green LEDs** to represent stop and crossing conditions. This differentiation is implemented to clearly distinguish between vehicle and pedestrian signals within the prototype, even though a complete real-world traffic system typically includes all three signal states (red, yellow, and green) for vehicles.



8. Implementation

The implementation of the system is divided into two main software modules.

The first module runs on the laptop and performs video capture, object detection, and decision making. This module uses the YOLOv8 model to detect objects and determine traffic conditions. After processing each frame, the module sends the decision message to the Raspberry Pi using socket communication.

The second module runs on the Raspberry Pi and functions as a server. It listens for incoming messages and interprets the commands received from the detection system.

Based on the command, the Raspberry Pi activates specific GPIO pins to control the LEDs.

This modular design allows the system to be flexible and scalable. The detection module can be upgraded or modified independently without affecting the hardware control module.

9. Results and Observations

The system successfully detects pedestrians and vehicles in real time using the YOLO object detection model. The detection results are processed quickly enough to allow dynamic traffic signal adjustments.

The Raspberry Pi receives commands from the detection system and accurately controls the LED signals representing traffic lights. The signals switch correctly depending on the detected objects and priority conditions.

The emergency vehicle priority feature ensures that traffic signals immediately allow vehicle passage when such vehicles are detected.

Overall, the system demonstrates reliable communication between the detection system and the hardware controller and effectively simulates a smart traffic management system.

10. Advantages

The proposed system improves pedestrian safety by allowing intelligent control of crossing signals based on real-time conditions. It reduces unnecessary waiting time for vehicles by avoiding fixed signal timing.

The system is cost-effective because it uses affordable hardware components such as Raspberry Pi and open-source software tools. The modular design allows easy expansion and integration with other smart city systems.

Additionally, the use of artificial intelligence enables the system to adapt dynamically to changing traffic conditions.

11. Limitations

Although the system performs effectively under normal conditions, certain limitations exist. The detection accuracy may decrease in poor lighting conditions or heavy weather conditions such as rain or fog.

The emergency vehicle detection mechanism is currently based on general vehicle classification and may require further refinement for accurate identification of ambulances or fire trucks.

The prototype system is designed for demonstration purposes and uses LEDs instead of real traffic lights.

12. Future Scope

The system can be further improved by integrating high-resolution cameras and deploying the detection algorithm directly on edge devices such as Raspberry Pi with optimized AI models.

Future developments could include integrating sound-based emergency detection using siren recognition, implementing wireless communication networks for city-wide traffic coordination, and connecting the system to centralized traffic management platforms.

The system could also incorporate additional sensors such as infrared sensors, LiDAR, or radar to improve detection reliability under different environmental conditions.

13. Conclusion

The Smart Zebra Crossing System with Emergency Traffic Control demonstrates how computer vision and embedded systems can be combined to create an intelligent traffic management solution. By using real-time object detection and automated signal control, the system improves pedestrian safety and optimizes traffic flow.

The integration of YOLO-based detection, socket communication, and Raspberry Pi GPIO control provides a practical demonstration of smart traffic technology. The project highlights the potential of artificial intelligence in modern transportation systems and offers a scalable approach that can be adapted for smart city infrastructure.